

Course Code	Course Title	L	T	P	C
BITE311L	Human Computer Interaction	3	0	0	3
Pre-requisite	NIL	Syllabus version			
		1.0			
Course Objectives:					
1. To learn the basic physiological, perceptual, and cognitive components of human learning and memory 2. To analyse interaction problems from a technical, cognitive and functional perspective 3. To gain practical experience in the fundamental aspects of designing, implementing and user interfaces					
Course Outcomes:					
1. Apply the principles and guidelines of human-computer interaction that must be considered when designing interactive systems 2. Analyse the design process, theories, models and interaction types for the problem statement 3. Formulate suitable methodology for the problem statement using appropriate methods, procedures and tools for the various phases of product development life cycle process 4. Predict, assess, evaluate and recommend the appropriate design to the target users 5. Learn the current state of research and development in human-computer interaction and make an effective study on any computer-based application and present for the assessments as an individual or team					
Module:1	Human Computer Interaction	6 hours			
Introduction - Good and Poor Design - Interaction Design - The User Experience - Understanding Users - Accessibility and Inclusiveness - Usability goals - User Experience Goals - Design principles.					
Module:2	Interaction Design Process and Conceptualizing Interaction	6 hours			
Introduction - Design process - Conceptualizing Interaction - Conceptual Models - Interface Metaphors - Interaction Types – Paradigms – Visions – Theories - Models and Frameworks.					
Module:3	Cognitive Aspects	6 hours			
Cognition - Kinds of Cognitive Processes – Attention – Perception – Memory – Learning – Reading - Speaking and Listening and Problem-Solving – Planning - Reasoning and Decision-Making - Cognitive Frameworks - Mental Models - Gulfs of Execution and Evaluation - Information Processing - Distributed Cognition - External Cognition - Embodied Interaction.					
Module:4	Social and Emotional Interaction	6 hours			

Introduction - Face-to-Face Conversations - Remote Conversations - Co-presence - Social Engagement - Emotions and the User Experience - Expressive and Annoying Interfaces - Affective Computing and Emotional AI - Persuasive Technologies and Behavior Change – Anthropomorphism.			
Module:5	Discovering Requirements		6 hours
Interfaces Types - Data Gathering: key Issues, Types: Analysis, Interpretation, Presentation: Types and Tools - Ethical Design Concerns - Data Gathering for Requirements – Personas - Capturing Interaction with Use Cases.			
Module:6	Interaction Design Process		7 hours
Introduction, Prototyping: What and Why Prototyping – Low-fidelity Types – High-fidelity Types - Compromising in Prototyping - Conceptual Design - Concrete Design - Generating Prototypes – Construction.			
Module:7	Evaluation		6 hours
Introduction: why, what, where, and when evaluation - Types of Evaluation – Usability Testing: Methods, Tasks and Users - Labs and Equipment - Conducting Experiments - Heuristic Evaluation - Walk throughs - Analytics in evaluation: A/B Testing - Fitt’s law.			
Module:8	Contemporary Issues		2 hours
	Total Lecture hours:		45 hours
Text Book			
1.	Helen Sharp, Yvonne Rogers, Jennifer Preece, Interaction Design: beyond human-computer interaction, 2019, Fifth Edition, Wiley.		
Reference Books			
1.	Gerard Jounghyun Kim, Human Computer Interaction – Fundamentals and Practice, – CRC press, 2015.		
2.	Ben Shneiderman, Catherine Plaisant, Maxine Cohen, Steven Jacobs, Designing the User Interface: Strategies for Effective Human Computer Interaction, 5th Edition, Pearson, 2009.		
3.	Alan Dix, Janet E. Finlay, Gregory D. Abowd, Russell Beale, Human - Computer Interaction, 3rd Edition, Pearson, 2003.		
Mode of Evaluation: Continuous Assessment Tests, Assignment, Quiz, Final Assessment Test			
Recommended by Board of Studies		12-10-2022	
Approved by Academic Council		No. 68	Date 19-12-2022